

PORTERVILLE, South Africa

WMO: 687170

Lat: 33.013S

Lon: 18.977E

Elev: 400

StdP: 14.48

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	37.3	39.3	33.0	27.9	53.9	35.0	30.2	51.4	16.6	54.5	15.0	55.5	1.6	90	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	98.3	68.5	95.0	67.5	91.8	66.7	70.8	92.7	69.3	89.8	68.1	87.6	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.7	89.5	76.4	62.3	85.3	74.8	61.2	81.8	74.5	34.8	92.3	33.5	90.2	32.6	87.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.8	11.7	34.1	106.2	1.6	1.5	33.0	107.3	32.0	108.2	31.1	109.0	30.0	110.1		
(5)			32.9	73.4	1.3	1.1	31.9	74.2	31.2	74.8	30.4	75.5	29.5	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	77.2	77.1	73.4	67.0	60.2	55.1	53.5	53.4	57.4	64.3	68.9	74.1		
	DBStd	10.23	5.67	5.87	5.77	6.23	5.03	4.36	4.57	3.87	4.79	6.17	5.97	5.57		
	HDD50	32	0	0	0	0	0	4	15	11	2	0	0	0		
	HDD65	1595	1	1	4	47	168	300	358	360	235	88	30	3		
	CDD50	5528	842	758	724	511	316	157	124	116	223	443	567	747		
	CDD65	1615	378	339	264	108	19	3	1	0	5	66	147	285		
	CDH74	15391	3794	3297	2190	896	144	25	17	9	110	775	1389	2745		
	CDH80	6902	1896	1629	926	311	24	3	1	0	32	284	551	1245		
(14) Wind	WSAvg	3.8	4.5	4.0	3.7	3.4	3.8	4.1	3.8	3.6	3.2	3.6	4.0	4.5		
(15) Precipitation	PrecAvg	17.4	0.4	0.4	0.6	1.3	2.4	3.0	2.7	2.7	1.4	0.9	0.9	0.6		
	PrecMax	25.4	1.9	2.8	2.4	2.6	5.5	8.1	6.9	5.4	3.7	2.4	2.2	2.0		
	PrecMin	11.6	0.0	0.0	0.0	0.2	0.2	0.4	0.9	1.0	0.3	0.1	0.0	0.0		
	PrecStd	3.7	0.4	0.7	0.6	0.7	1.5	1.6	1.6	1.1	1.0	0.6	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.3	101.3	99.0	93.7	83.3	78.0	76.2	74.9	83.1	95.2	96.5	99.8		
		MCWB	70.1	69.8	68.6	65.6	61.5	59.7	61.0	61.3	63.1	66.5	66.6	68.8		
	2%	DB	98.2	97.0	93.0	87.8	78.1	71.0	70.0	69.1	76.5	87.6	90.9	94.7		
		MCWB	69.1	68.4	66.9	64.8	60.0	57.0	58.2	58.5	60.5	64.1	64.8	66.7		
	5%	DB	94.2	93.4	89.0	83.3	74.0	66.4	65.5	64.8	71.7	82.5	86.5	91.0		
		MCWB	67.9	67.7	65.7	63.1	58.9	56.2	56.0	56.1	58.4	61.9	63.5	66.0		
	10%	DB	90.0	89.6	85.2	79.1	69.5	62.6	61.8	61.3	67.3	77.7	82.4	87.2		
		MCWB	66.7	66.9	64.9	62.2	57.8	54.8	54.2	54.0	56.8	60.2	62.2	65.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	72.8	70.8	68.0	64.0	61.0	62.6	63.0	65.1	68.0	68.6	70.5		
		MCDB	96.7	94.8	92.5	88.3	76.6	71.9	73.2	72.6	80.0	90.8	89.4	95.0		
	2%	WB	70.7	70.8	68.6	66.2	61.7	58.5	59.1	59.3	61.7	65.4	66.4	68.4		
		MCDB	93.8	91.0	88.8	82.8	73.8	66.9	69.0	67.8	74.5	84.1	85.8	90.1		
	5%	WB	69.0	69.4	67.1	64.7	60.2	57.1	56.7	56.6	59.4	63.1	64.8	67.1		
		MCDB	89.8	88.3	85.4	79.7	70.0	63.5	63.8	63.5	70.4	79.0	82.9	87.1		
	10%	WB	67.7	67.9	65.7	62.9	58.7	55.7	54.8	54.5	57.4	61.1	63.2	65.7		
		MCDB	86.6	85.9	82.0	76.9	67.5	61.0	60.5	60.2	66.2	74.7	79.2	83.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.7	24.2	23.1	21.9	17.8	16.8	18.2	17.7	20.6	24.6	24.6	24.7		
		MCDBR	30.5	29.3	28.5	28.7	26.3	24.5	25.4	25.6	29.5	33.3	32.3	30.6		
		MCWBR	8.8	8.2	8.8	10.1	11.9	13.9	16.2	16.2	14.9	13.1	10.7	9.5		
	5% WB	MCDBR	27.7	25.1	25.8	24.7	22.1	19.4	22.6	23.5	27.0	30.7	28.9	27.8		
		MCWBR	8.6	7.7	8.6	9.5	10.8	12.0	14.9	15.4	14.6	12.6	10.2	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.337	0.333	0.318	0.310	0.303	0.306	0.330	0.341	0.352	0.332	0.336		
	taud		2.481	2.483	2.485	2.534	2.548	2.542	2.523	2.449	2.387	2.373	2.471	2.477		
	Ebn at Noon		318	311	301	289	274	268	272	279	293	302	317	319		
	Edh at Noon		37	36	33	28	25	23	25	30	36	39	37	37		
(44) All-Sky Solar Radiation	RadAvg		2666	2406	1958	1436	1018	858	957	1194	1640	2126	2507	2694		
	RadStd		69	62	60	73	67	53	55	56	69	101	74	88		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		+0.63	N/A	N/A	+0.38	N/A	N/A	-4	-133	+200	+110	

Nomenclature: See separate page